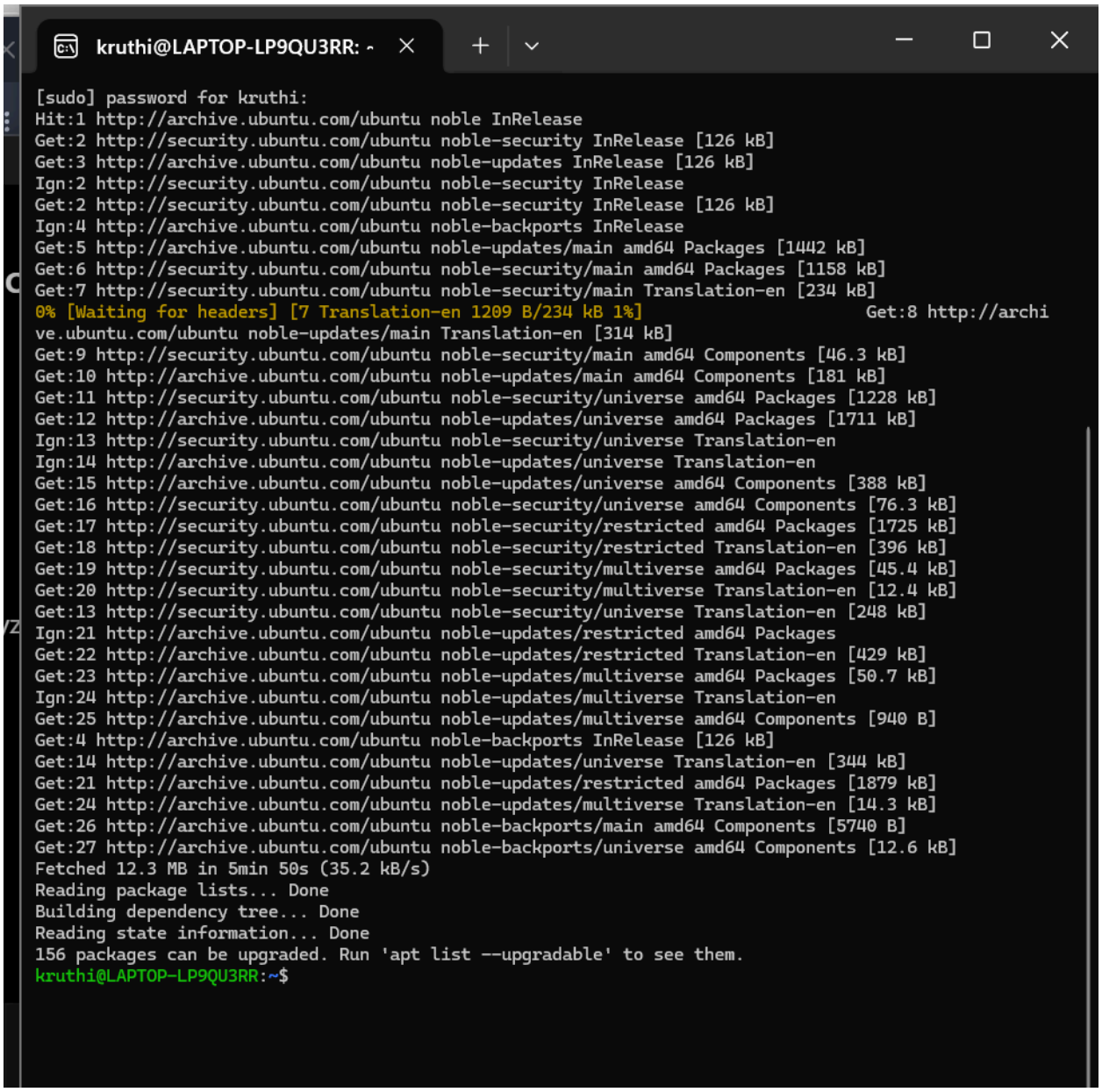
Cnnp skill-2

THREE-POINT CAPTURE – Analyze packet captures from three points in a network path to understand packet flow and identify performance differences.

Stage 1 Install tools



STAGE 2 — Build the network

Open again ubuntu and rename that tab as Terminal 1 and run all of these one by one:

sudo ip netns add host1 sudo ip netns add router sudo ip netns add host2

```
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 883 (unattended-upgr)
```

```
Setting up libqt6waylandclient6:amd64 (6.4.2-5build3) ...
Setting up qt6-gtk-platformtheme:amd64 (6.4.2+dfsg-21.1build5) ...
Setting up libqt6printsupport6t64:amd64 (6.4.2+dfsg-21.1build5) ...
Setting up libqt6quick6:amd64 (6.4.2+dfsg-4build3) ...
Setting up libqt6wlshellintegration6:amd64 (6.4.2-5build3) ...
Setting up wireshark (4.2.2-1.1build3) ...
Setting up libqt6waylandcompositor6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandeglcompositorhwintegration6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandeglclienthwintegration6:amd64 (6.4.2-5build3) ...
Setting up qt6-wayland:amd64 (6.4.2-5build3) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for shared-mime-info (2.4-4) ...
Processing triggers for udev (255.4-1ubuntu8.17) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add host2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns list
host2
router
host1
kruthi@LAPTOP-LP9QU3RR:~$
```

```
Setting up libqt6waylandeglcompositorhwintegration6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandeglclienthwintegration6:amd64 (6.4.2-5build3) ...
Setting up qt6-wayland:amd64 (6.4.2-5build3) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for shared-mime-info (2.4-4) ...
Processing triggers for udev (255.4-1ubuntu8.17) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add host2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns list
host2
router
host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-h1 type veth peer name veth-r1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-r2 type veth peer name veth-h2
kruthi@LAPTOP-LP9QU3RR:~$
```

```
Setting up libqt6wlshellintegration6:amd64 (6.4.2-5build3) ...
Setting up wireshark (4.2.2-1.1build3) ...
Setting up libqt6waylandcompositor6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandeglcompositorhwintegration6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandeglclienthwintegration6:amd64 (6.4.2-5build3) ...
Setting up qt6-wayland:amd64 (6.4.2-5build3) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for shared-mime-info (2.4-4) ...
Processing triggers for udev (255.4-1ubuntu8.17) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add host2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns list
host2
router
host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-h1 type veth peer name veth-r1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-r2 type veth peer name veth-h2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h1 netns host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r1 netns router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r2 netns router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h2 netns host2
kruthi@LAPTOP-LP9QU3RR:~$
```

```
Setting up libqt6wlshellintegration6:amd64 (6.4.2-5build3) ...
Setting up wireshark (4.2.2-1.1build3) ...
Setting up libqt6waylandcompositor6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandeglcompositorhwintegration6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandeglclienthwintegration6:amd64 (6.4.2-5build3) ...
Setting up qt6-wayland:amd64 (6.4.2-5build3) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for shared-mime-info (2.4-4) ...
Processing triggers for udev (255.4-1ubuntu8.17) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add host2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns list
host2
router
host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-h1 type veth peer name veth-r1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-r2 type veth peer name veth-h2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h1 netns host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r1 netns router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r2 netns router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h2 netns host2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set veth-h1 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$
```

Stage - 3

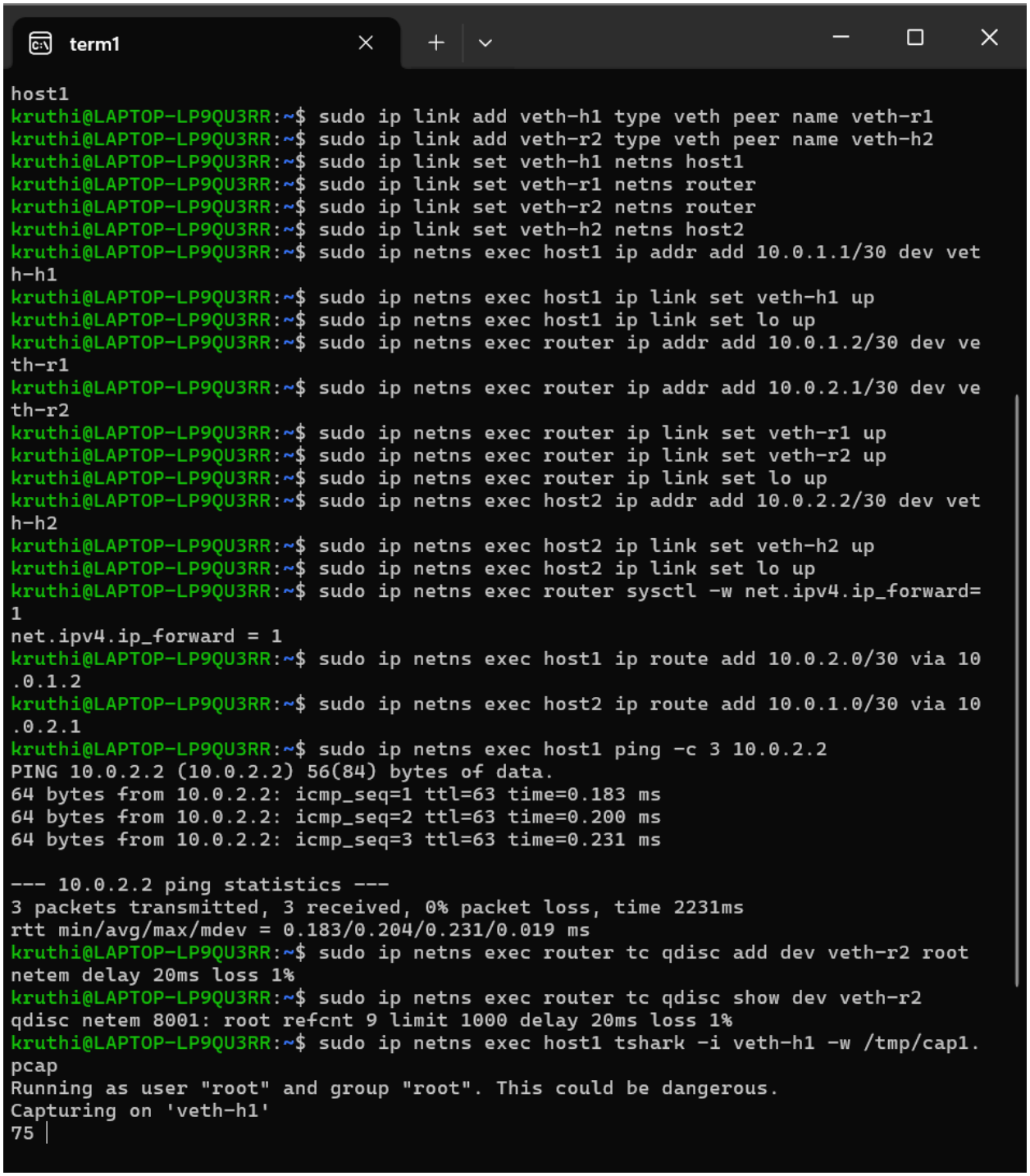
```
host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-h1 type veth peer name veth-r1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-r2 type veth peer name veth-h2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h1 netns host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r1 netns router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r2 netns router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h2 netns host2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set veth-h1 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set veth-r1 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set veth-r2 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$
```

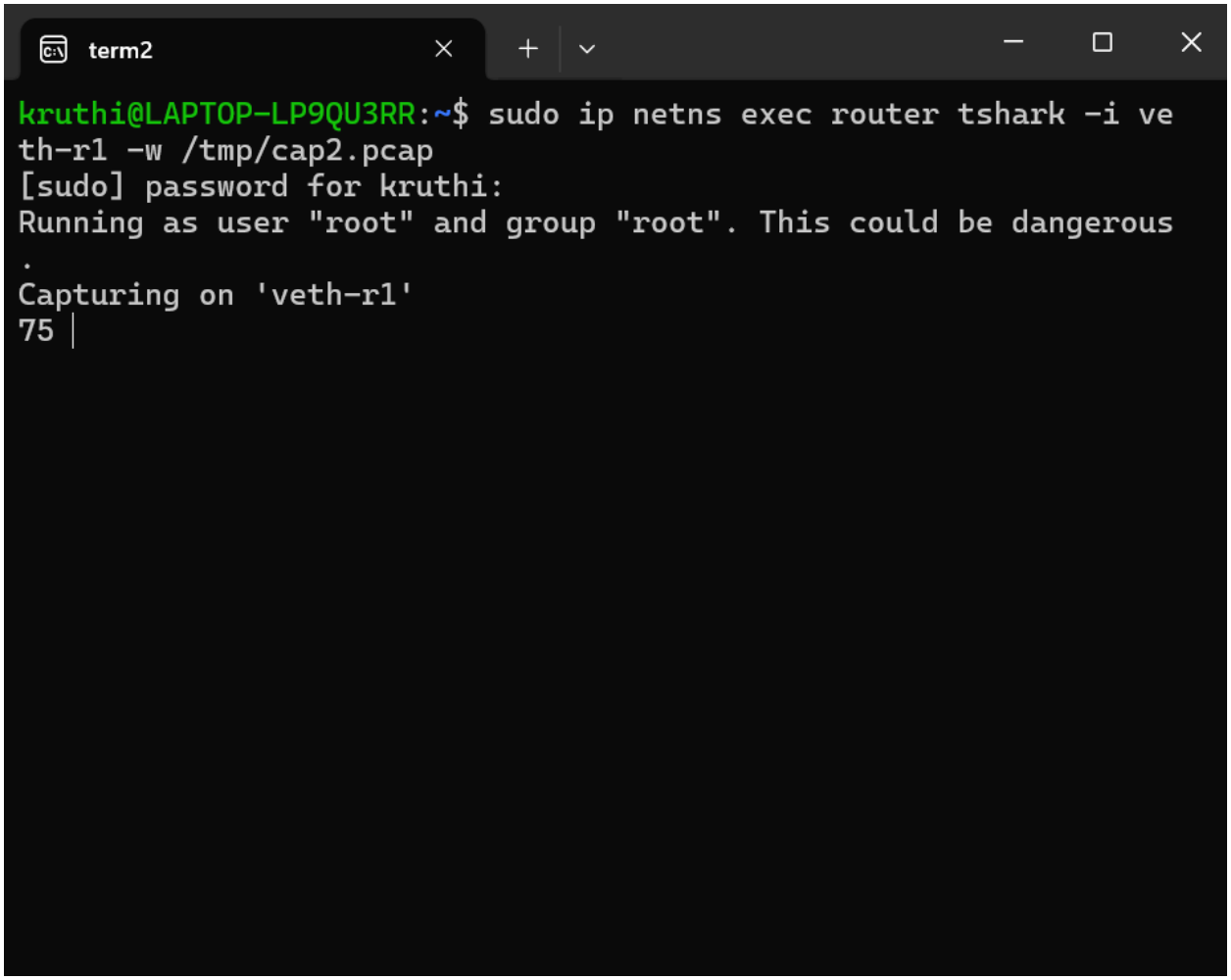
```
host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-h1 type veth peer name veth-r1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-r2 type veth peer name veth-h2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h1 netns host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r1 netns router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r2 netns router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h2 netns host2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set veth-h1 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set veth-r1 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set veth-r2 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip link set veth-h2 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$
```

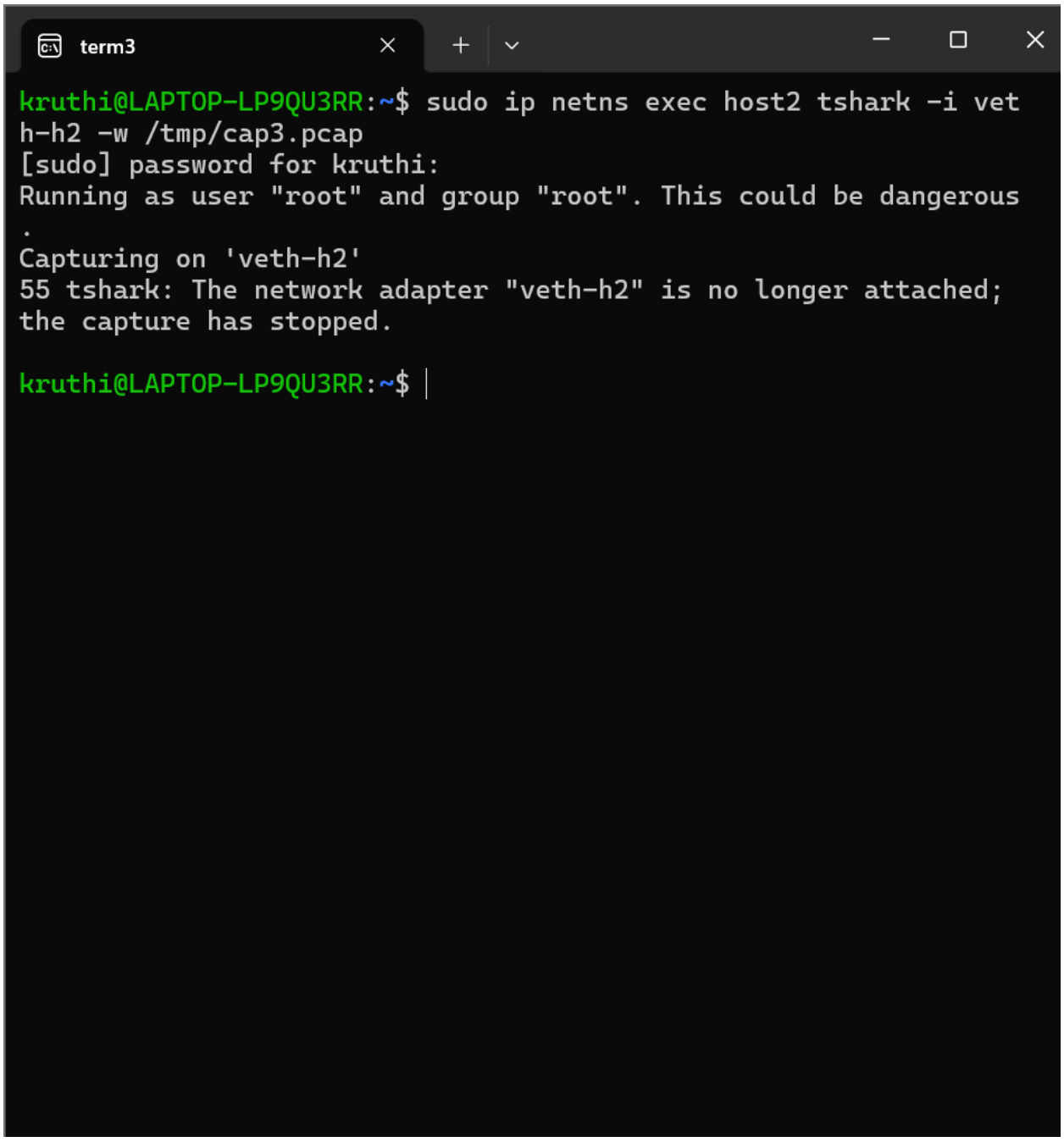
```
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set veth-h1 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set veth-r1 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set veth-r2 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip link set veth-h2 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
kruthi@LAPTOP-LP9QU3RR:~$
```

```
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=1.25 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.045 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.038 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2196ms
rtt min/avg/max/mdev = 0.038/0.443/1.247/0.568 ms
kruthi@LAPTOP-LP9QU3RR:~$
```


term1
host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-h1 type veth peer name veth-r1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-r2 type veth peer name veth-h2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h1 netns host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r1 netns router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r2 netns router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h2 netns host2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set veth-h1 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set veth-r1 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set veth-r2 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip link set veth-h2 up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip link set lo up
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.183 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.200 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.231 ms
--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2231ms
rtt min/avg/max/mdev = 0.183/0.204/0.231/0.019 ms
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 9 limit 1000 delay 20ms loss 1%
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
75

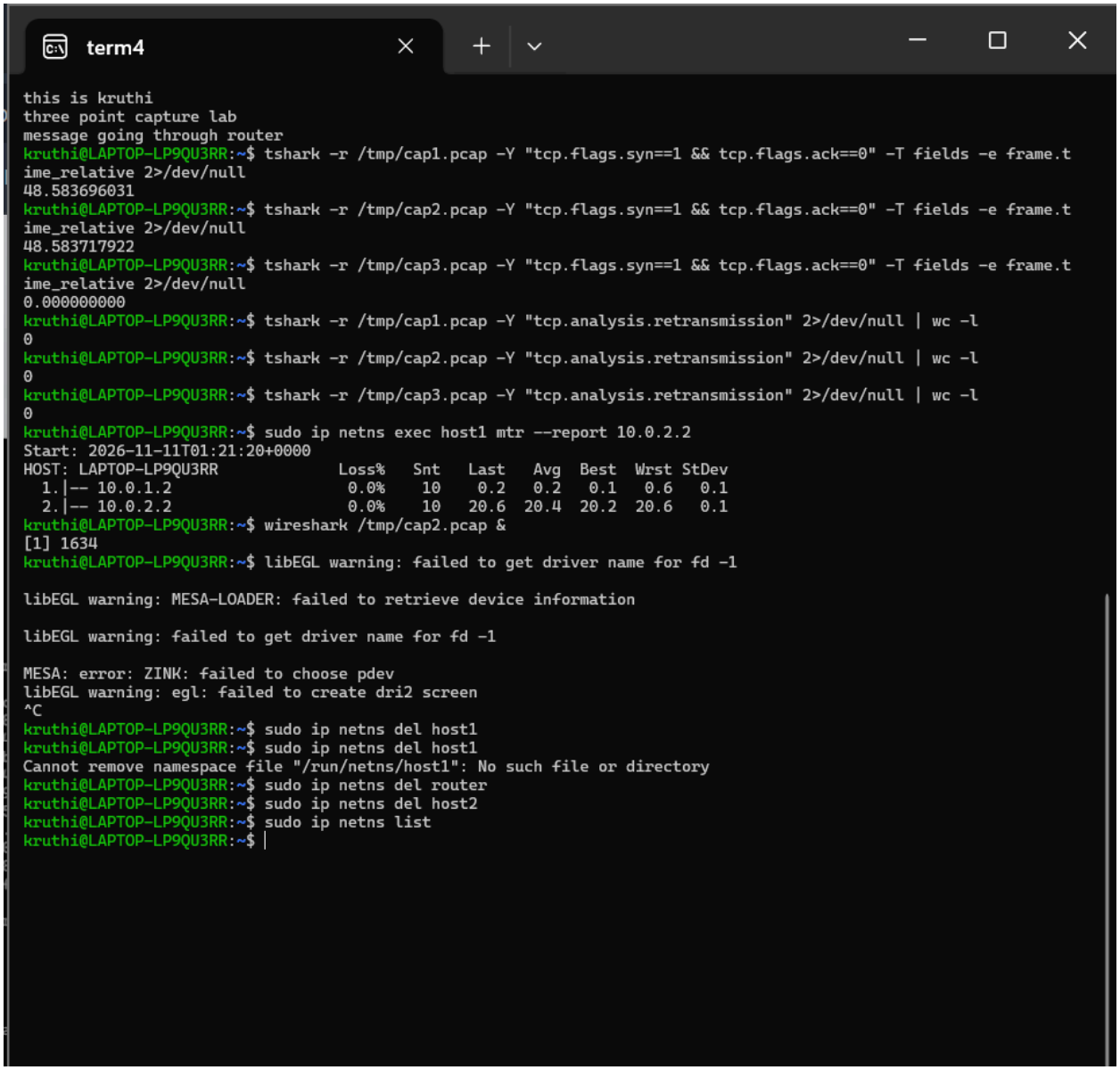

term2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo] password for kruthi:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
75

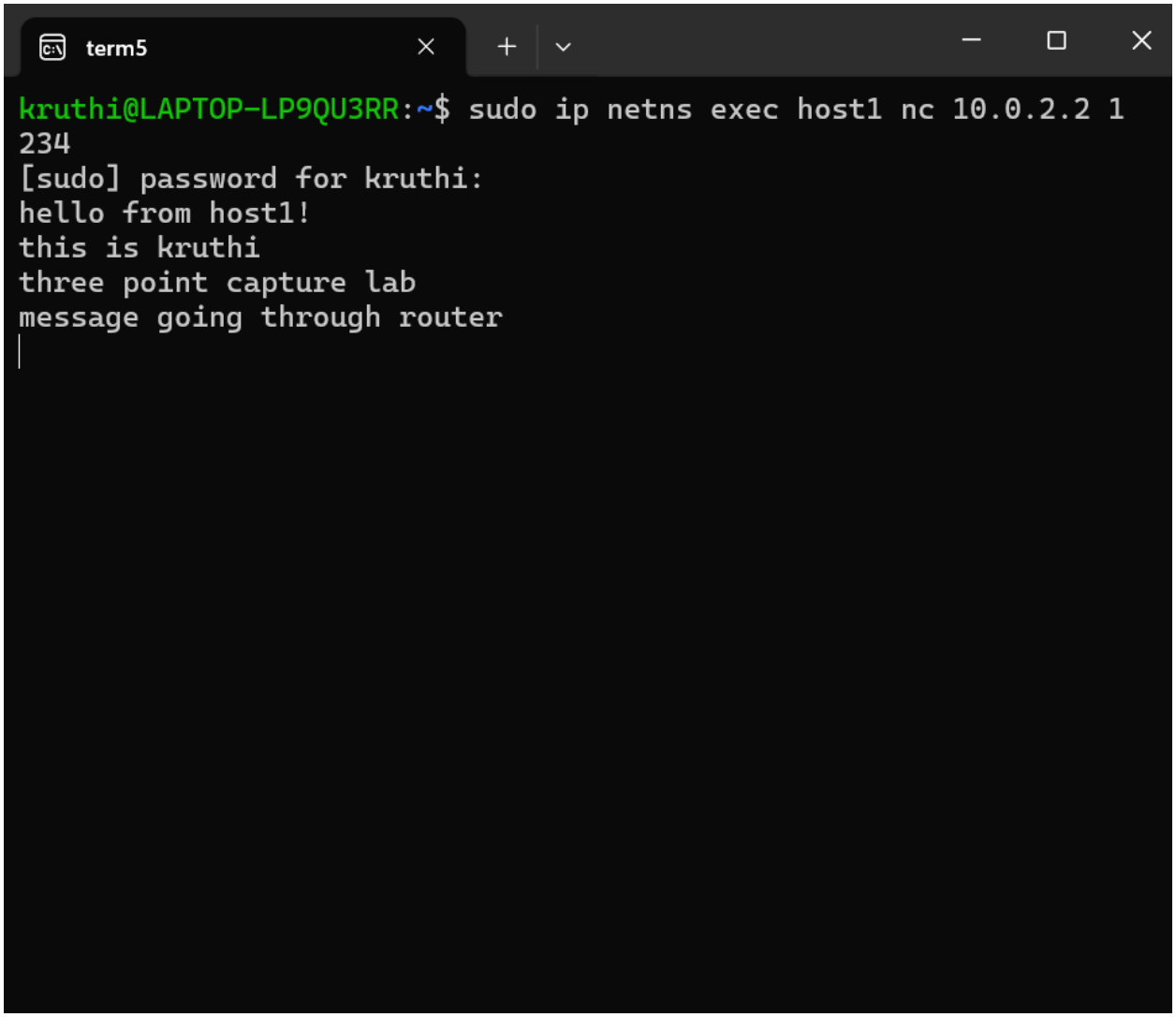

term3
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo] password for kruthi:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
55 tshark: The network adapter "veth-h2" is no longer attached; the capture has stopped.
kruthi@LAPTOP-LP9QU3RR:~$

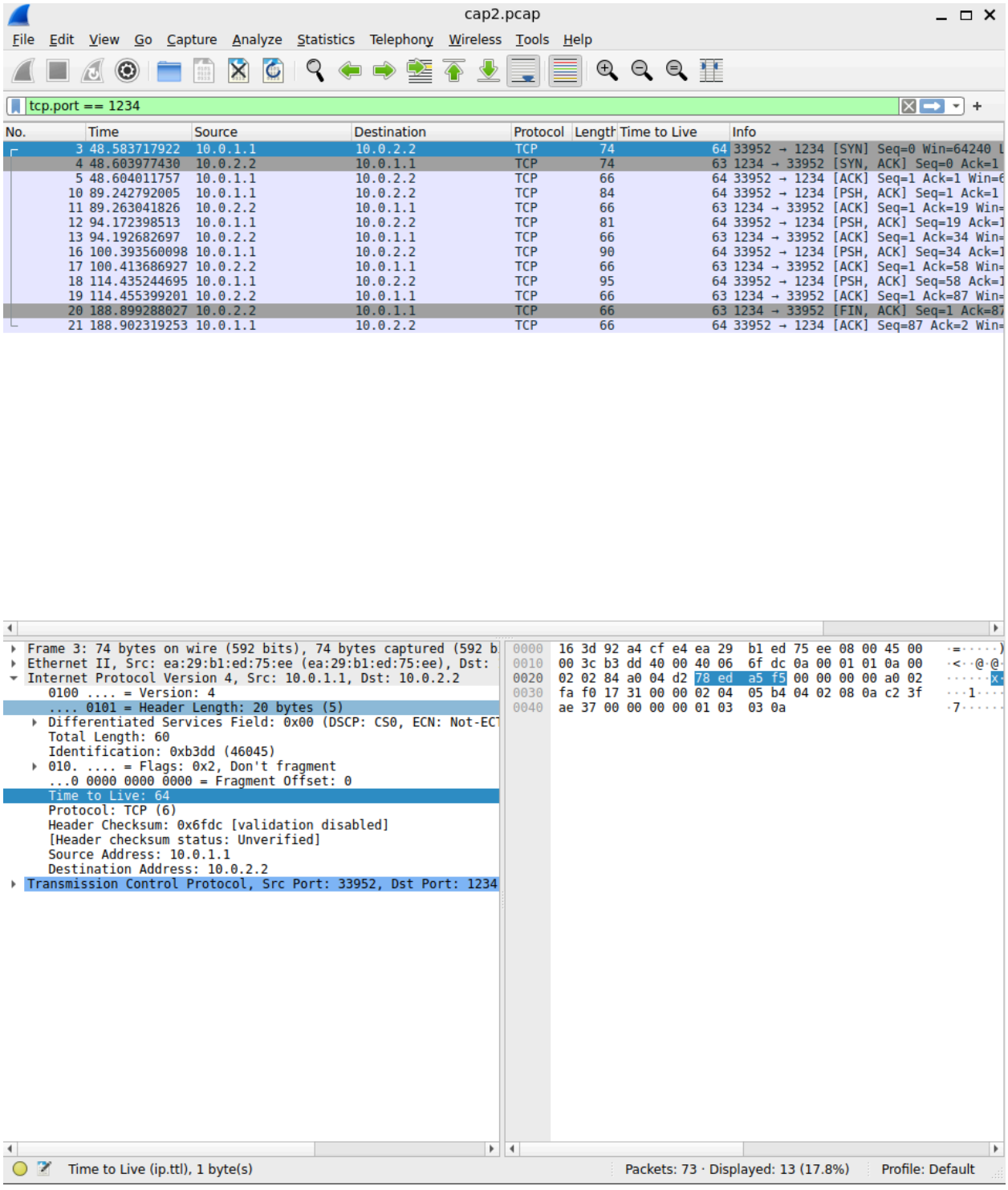
```
term4
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 nc -l 1234
[sudo] password for kruthi:
hello from host1!
this is kruthi
three point capture lab
message going through router
^C
kruthi@LAPTOP-LP9QU3RR:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
kruthi@LAPTOP-LP9QU3RR:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 2.6K Nov 11 01:17 /tmp/cap1.pcap
-rw-r--r-- 1 root root 2.6K Nov 11 01:17 /tmp/cap2.pcap
-rw-r--r-- 1 root root 2.6K Nov 11 01:17 /tmp/cap3.pcap
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      6 63
      7 64
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      6 63
      7 64
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      7 63
      6 64
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e data.text 2>/dev/null

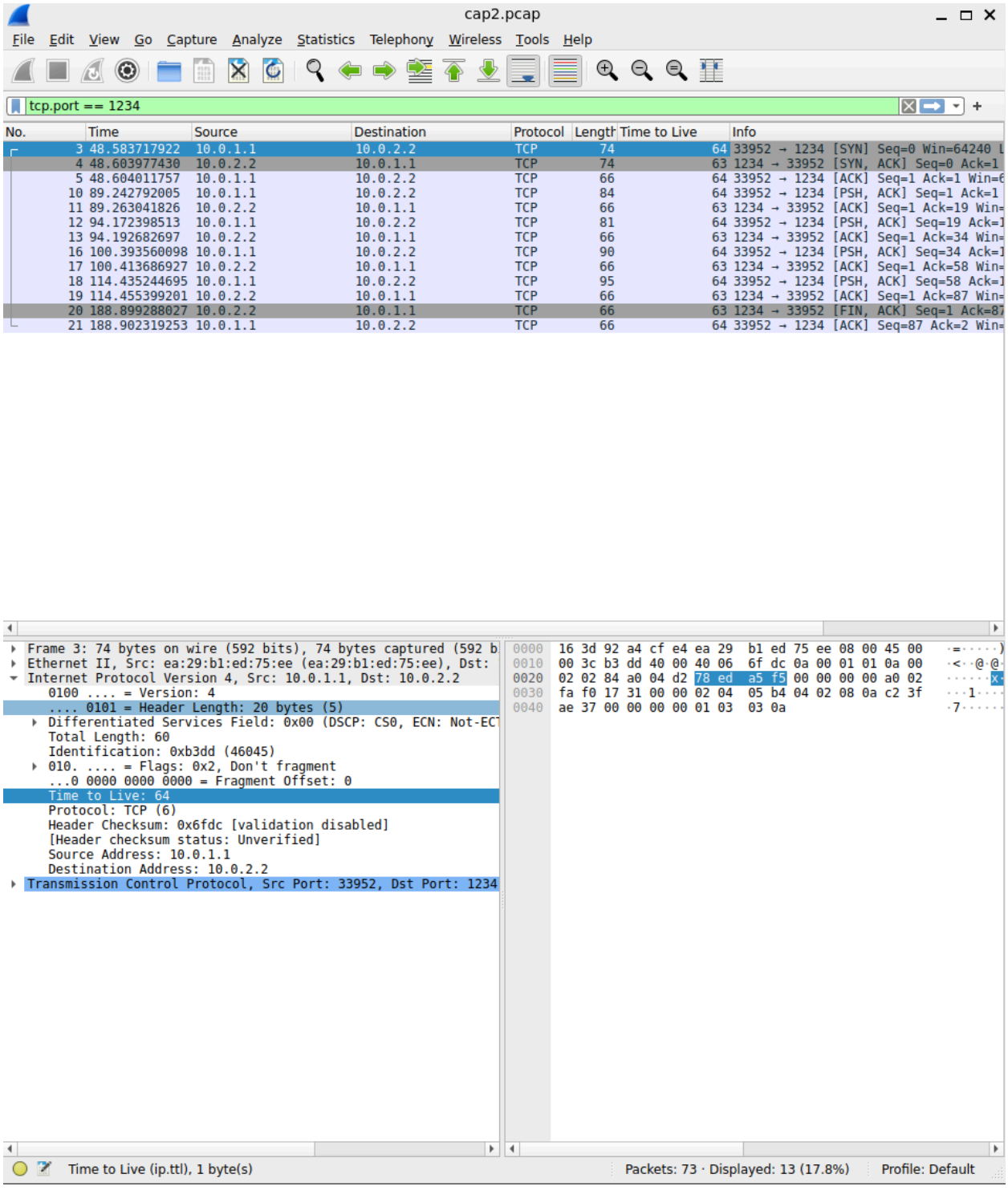


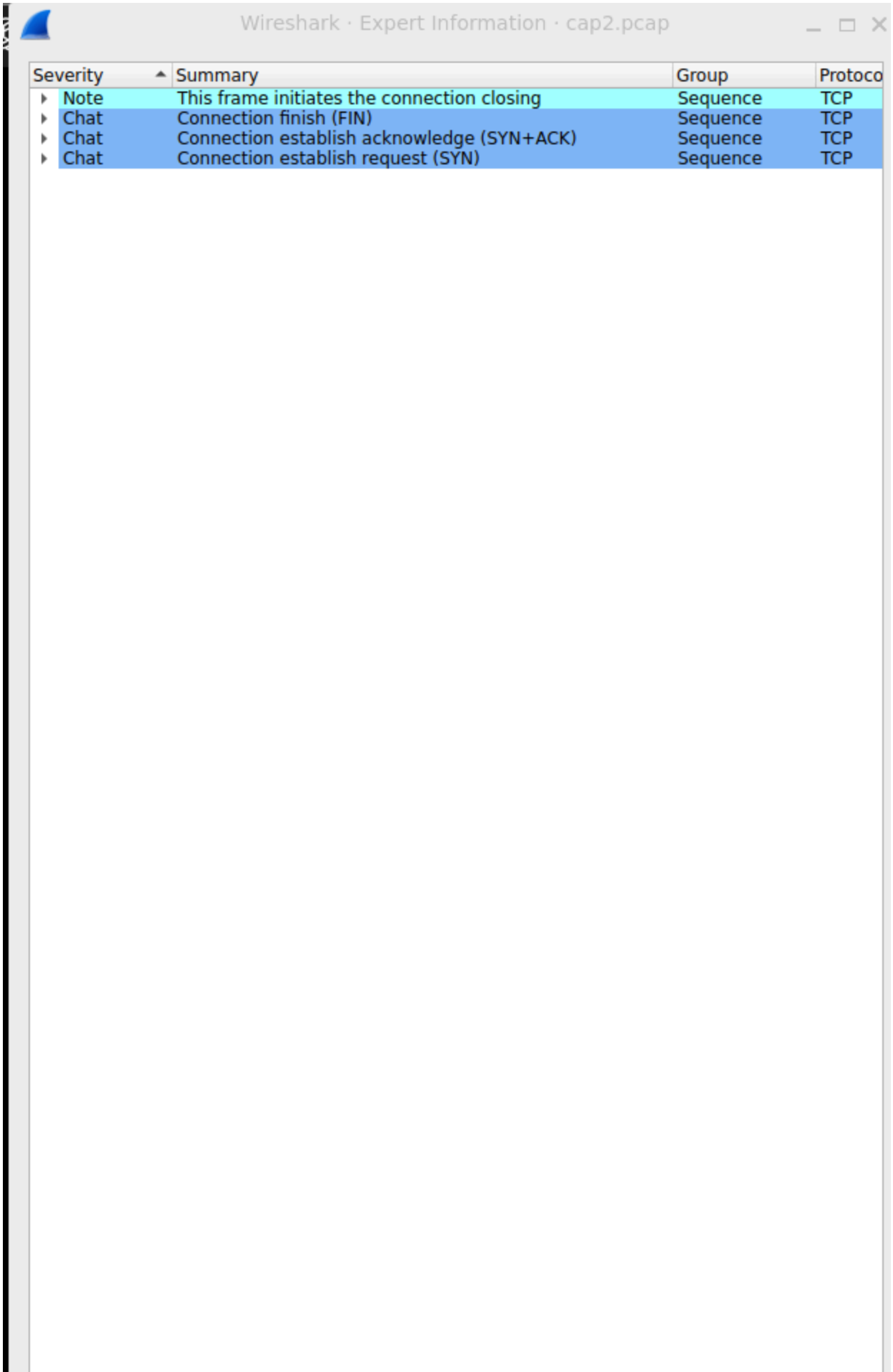

kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload 2>/dev/null | x
xd | head -20
00000000: 0a0a 0a36 3836 3536 6336 6336 6632 3036  ...68656c6c6f206
00000010: 3637 3236 6636 6432 3036 3836 6637 3337  6726f6d20686f737
00000020: 3433 3132 3130 610a 0a37 3436 3836 3937  431210a..7468697
00000030: 3332 3036 3937 3332 3036 6237 3237 3537  3206973206b72757
00000040: 3436 3836 3930 610a 0a37 3436 3837 3236  468690a..7468726
00000050: 3536 3532 3037 3036 6636 3936 6537 3432  56520706f696e742
00000060: 3036 3336 3137 3037 3437 3537 3236 3532  0636170747572652
00000070: 3036 6336 3136 3230 610a 0a36 6436 3537  06c61620a..6d657
00000080: 3337 3336 3136 3736 3532 3036 3736 6636  3736167652067 6f6
00000090: 3936 6536 3732 3037 3436 3837 3236 6637  96e67207468726f7
000000a0: 3536 3736 3832 3037 3236 6637 3537 3436  5676820726f75746
000000b0: 3537 3230 610a 0a0a 0a                   5720a....
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload 2>/dev/null | t
r -d ':' | xxd -r -p
hello from host1!
this is kruthi
three point capture lab
message going through router
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.t
ime_relative 2>/dev/null
48.583696031
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.t
ime_relative 2>/dev/null
48.583717922
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.t
ime_relative 2>/dev/null
0.000000000
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
```

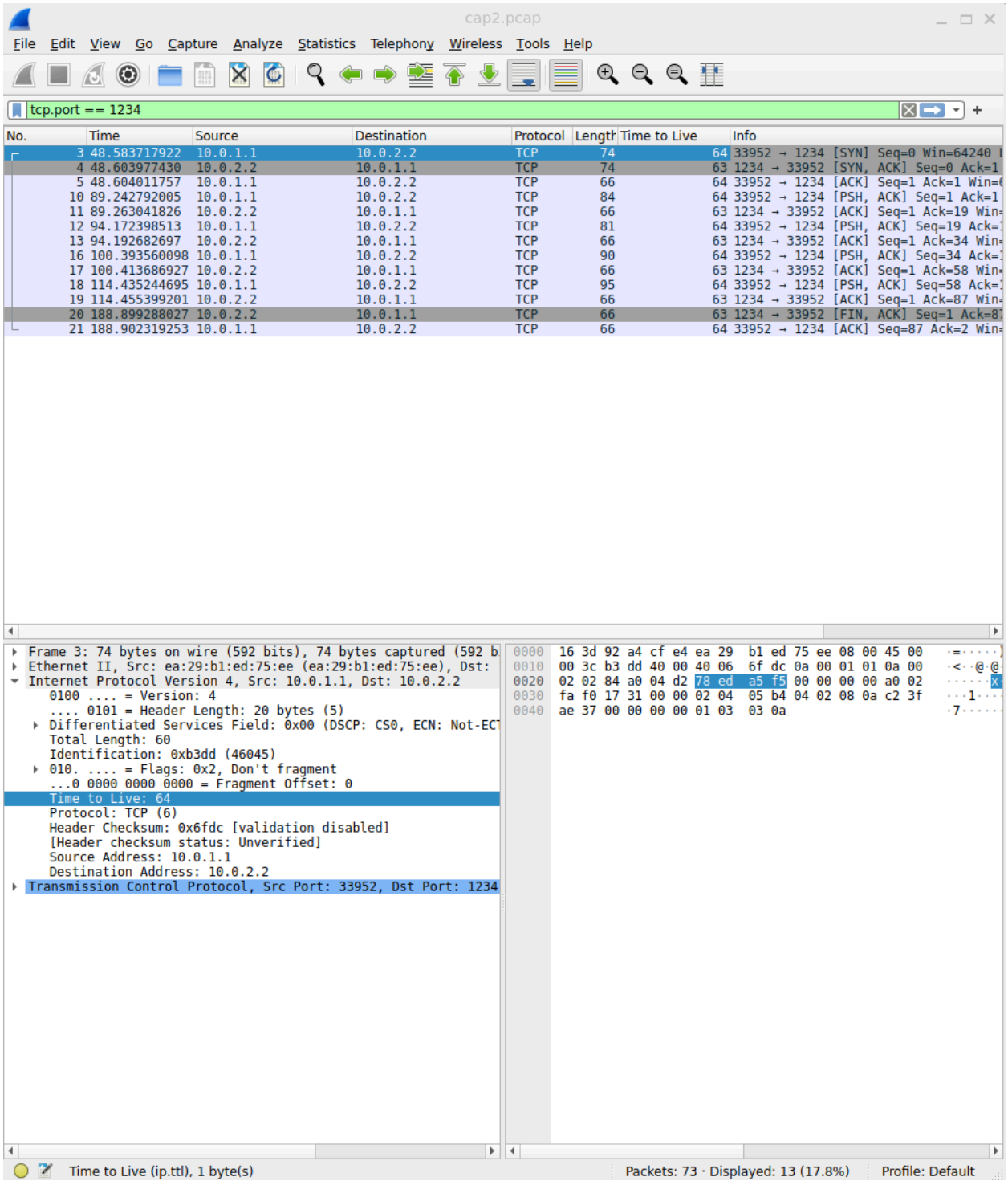

term4
this is kruthi
three point capture lab
message going through router
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
48.583696031
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
48.583717922
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T fields -e frame.time_relative 2>/dev/null
0.000000000
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
kruthi@LAPTOP-LP9QU3RR:~$ tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-11T01:21:20+0000
HOST: LAPTOP-LP9QU3RR Loss% Snt Last Avg Best Wrst StDev
1.|-- 10.0.1.2 0.0% 10 0.2 0.2 0.1 0.6 0.1
2.|-- 10.0.2.2 0.0% 10 20.6 20.4 20.2 20.6 0.1
kruthi@LAPTOP-LP9QU3RR:~$ wireshark /tmp/cap2.pcap &
[1] 1634
kruthi@LAPTOP-LP9QU3RR:~$ libEGL warning: failed to get driver name for fd -1
libEGL warning: MESA-LOADER: failed to retrieve device information
libEGL warning: failed to get driver name for fd -1
MESA: error: ZINK: failed to choose pdev
libEGL warning: egl: failed to create dri2 screen
^C
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns del host1
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns del host1
Cannot remove namespace file "/run/netns/host1": No such file or directory
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns del router
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns del host2
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns list
kruthi@LAPTOP-LP9QU3RR:~$

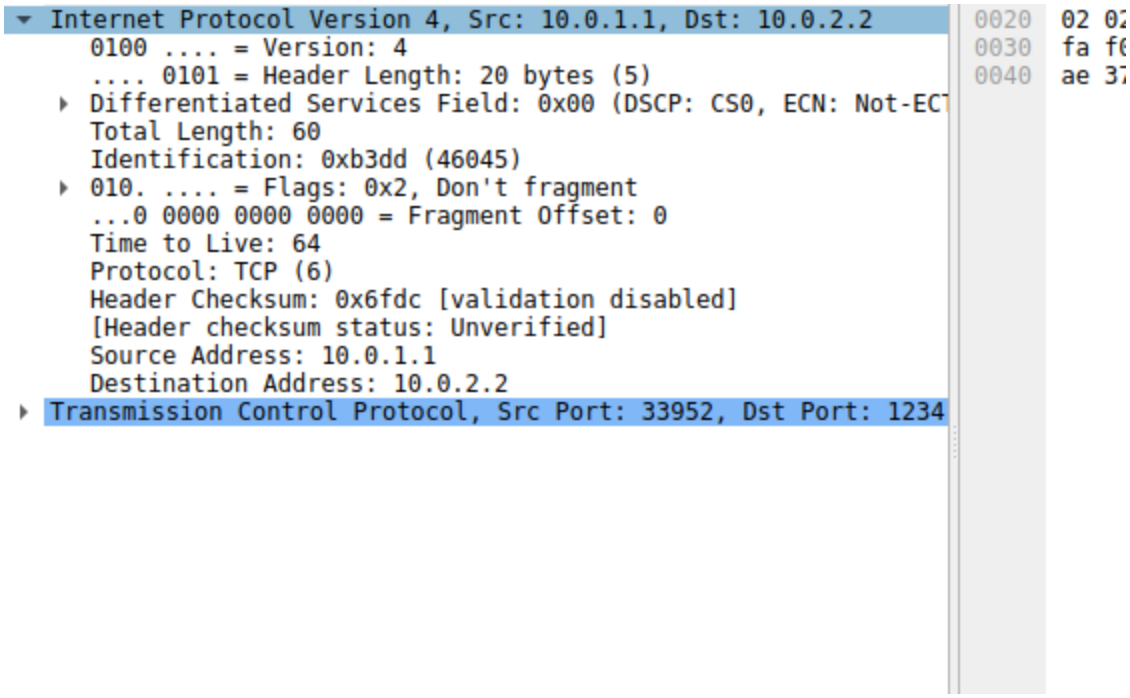

term5
kruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 nc 10.0.2.2 1
234
[sudo] password for kruthi:
hello from host1!
this is kruthi
three point capture lab
message going through router

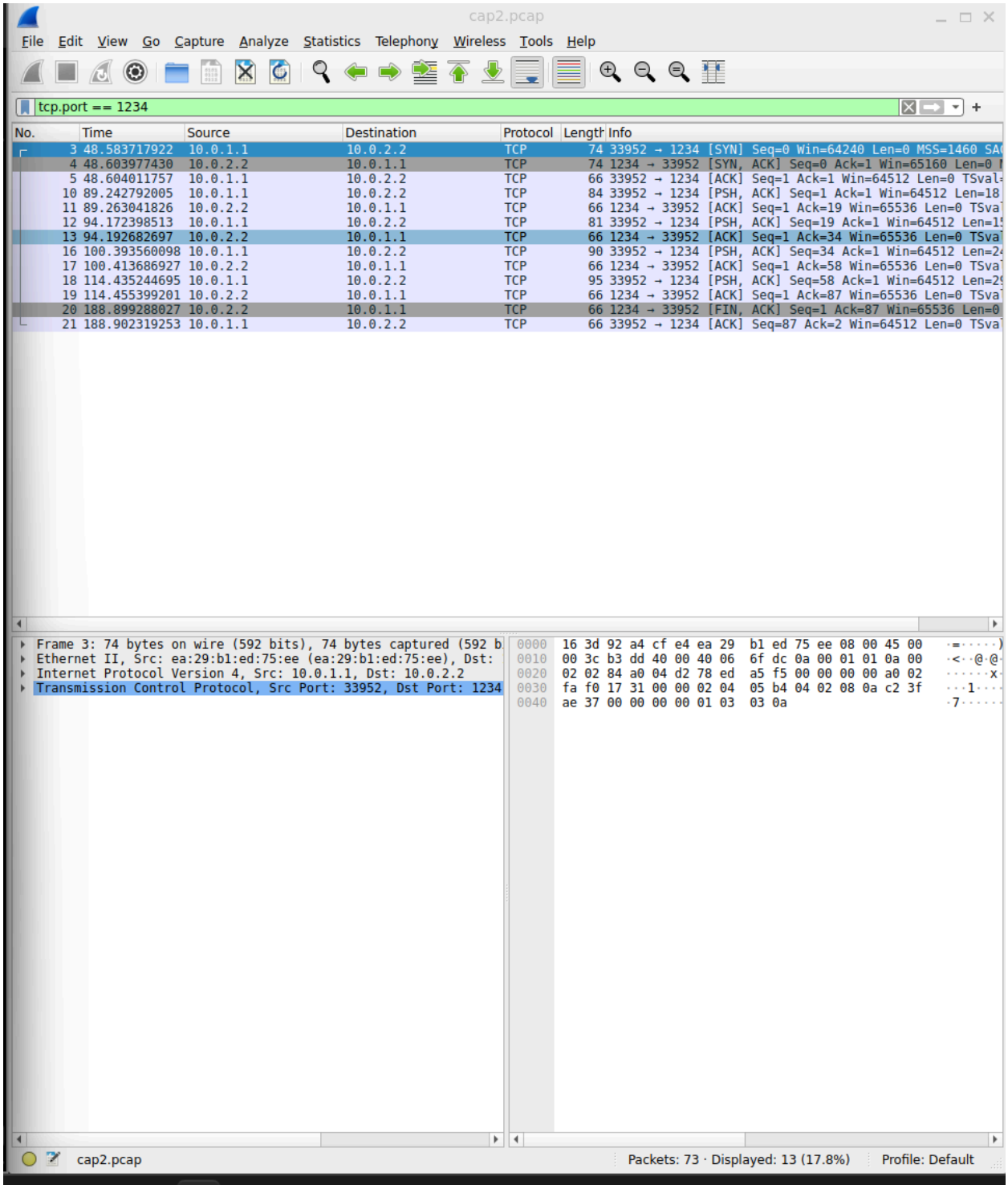

cap2.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 1234
No. Time Source Destination Protocol Length Time to Live Info
3 48.583717922 10.0.1.1 10.0.2.2 TCP 74 64 33952 → 1234 [SYN] Seq=0 Win=64240 L
4 48.603977430 10.0.2.2 10.0.1.1 TCP 74 63 1234 → 33952 [SYN, ACK] Seq=0 Ack=1
5 48.604011757 10.0.1.1 10.0.2.2 TCP 66 64 33952 → 1234 [ACK] Seq=1 Ack=1 Win=6
10 89.242792005 10.0.1.1 10.0.2.2 TCP 84 64 33952 → 1234 [PSH, ACK] Seq=1 Ack=1
11 89.263041826 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=19 Win=
12 94.172398513 10.0.1.1 10.0.2.2 TCP 81 64 33952 → 1234 [PSH, ACK] Seq=19 Ack=1
13 94.192682697 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=34 Win=
16 100.393560098 10.0.1.1 10.0.2.2 TCP 90 64 33952 → 1234 [PSH, ACK] Seq=34 Ack=1
17 100.413686927 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=58 Win=
18 114.435244695 10.0.1.1 10.0.2.2 TCP 95 64 33952 → 1234 [PSH, ACK] Seq=58 Ack=1
19 114.455399201 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=87 Win=
20 188.899288027 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [FIN, ACK] Seq=1 Ack=87
21 188.902319253 10.0.1.1 10.0.2.2 TCP 66 64 33952 → 1234 [ACK] Seq=87 Ack=2 Win=
Frame 3: 74 bytes on wire (592 bits), 74 bytes captured (592 b
Ethernet II, Src: ea:29:b1:ed:75:ee (ea:29:b1:ed:75:ee), Dst:
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT
Total Length: 60
Identification: 0xb3dd (46045)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x6fdc [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.1.1
Destination Address: 10.0.2.2
Transmission Control Protocol, Src Port: 33952, Dst Port: 1234
0000 16 3d 92 a4 cf e4 ea 29 b1 ed 75 ee 08 00 45 00
0010 00 3c b3 dd 40 00 40 06 6f dc 0a 00 01 01 0a 00
0020 02 02 84 a0 04 d2 78 ed a5 f5 00 00 00 00 a0 02
0030 fa f0 17 31 00 00 02 04 05 b4 04 02 08 0a c2 3f
0040 ae 37 00 00 00 00 01 03 03 0a
Time to Live (ip.ttl), 1 byte(s)
Packets: 73 · Displayed: 13 (17.8%)
Profile: Default

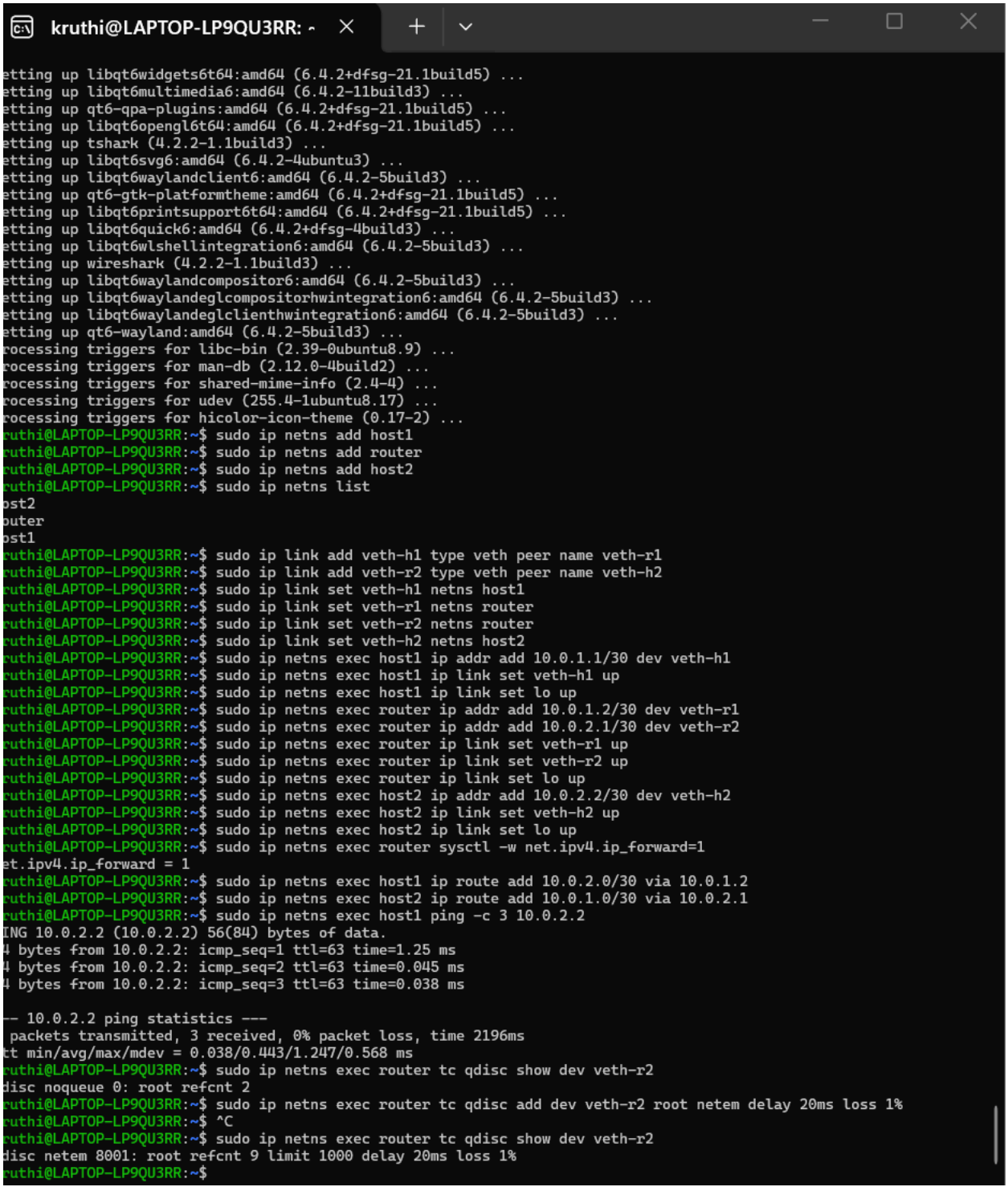

cap2.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 1234
No. Time Source Destination Protocol Length Time to Live Info
3 48.583717922 10.0.1.1 10.0.2.2 TCP 74 64 33952 → 1234 [SYN] Seq=0 Win=64240 L
4 48.603977430 10.0.2.2 10.0.1.1 TCP 74 63 1234 → 33952 [SYN, ACK] Seq=0 Ack=1
5 48.604011757 10.0.1.1 10.0.2.2 TCP 66 64 33952 → 1234 [ACK] Seq=1 Ack=1 Win=6
10 89.242792005 10.0.1.1 10.0.2.2 TCP 84 64 33952 → 1234 [PSH, ACK] Seq=1 Ack=1
11 89.263041826 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=19 Win=
12 94.172398513 10.0.1.1 10.0.2.2 TCP 81 64 33952 → 1234 [PSH, ACK] Seq=19 Ack=1
13 94.192682697 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=34 Win=
16 100.393560098 10.0.1.1 10.0.2.2 TCP 90 64 33952 → 1234 [PSH, ACK] Seq=34 Ack=1
17 100.413686927 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=58 Win=
18 114.435244695 10.0.1.1 10.0.2.2 TCP 95 64 33952 → 1234 [PSH, ACK] Seq=58 Ack=1
19 114.455399201 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=87 Win=
20 188.899288027 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [FIN, ACK] Seq=1 Ack=87
21 188.902319253 10.0.1.1 10.0.2.2 TCP 66 64 33952 → 1234 [ACK] Seq=87 Ack=2 Win=
Frame 3: 74 bytes on wire (592 bits), 74 bytes captured (592 b
Ethernet II, Src: ea:29:b1:ed:75:ee (ea:29:b1:ed:75:ee), Dst:
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT
Total Length: 60
Identification: 0xb3dd (46045)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x6fdc [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.1.1
Destination Address: 10.0.2.2
Transmission Control Protocol, Src Port: 33952, Dst Port: 1234
0000 16 3d 92 a4 cf e4 ea 29 b1 ed 75 ee 08 00 45 00
0010 00 3c b3 dd 40 00 40 06 6f dc 0a 00 01 01 0a 00
0020 02 02 84 a0 04 d2 78 ed a5 f5 00 00 00 00 a0 02
0030 fa f0 17 31 00 00 02 04 05 b4 04 02 08 0a c2 3f
0040 ae 37 00 00 00 00 01 03 03 0a
Time to Live (ip.ttl), 1 byte(s)
Packets: 73 · Displayed: 13 (17.8%)
Profile: Default


Wireshark · Expert Information · cap2.pcap
Severity
Summary
Group
Protoco
Note
This frame initiates the connection closing
Sequence
TCP
Chat
Connection finish (FIN)
Sequence
TCP
Chat
Connection establish acknowledge (SYN+ACK)
Sequence
TCP
Chat
Connection establish request (SYN)
Sequence
TCP


cap2.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 1234
No. Time Source Destination Protocol Length Time to Live Info
3 48.583717922 10.0.1.1 10.0.2.2 TCP 74 64 33952 → 1234 [SYN] Seq=0 Win=64240
4 48.603977430 10.0.2.2 10.0.1.1 TCP 74 63 1234 → 33952 [SYN, ACK] Seq=0 Ack=1
5 48.604011757 10.0.1.1 10.0.2.2 TCP 66 64 33952 → 1234 [ACK] Seq=1 Ack=1 Win=
10 89.242792005 10.0.1.1 10.0.2.2 TCP 84 64 33952 → 1234 [PSH, ACK] Seq=1 Ack=1
11 89.263041826 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=19 Win=
12 94.172398513 10.0.1.1 10.0.2.2 TCP 81 64 33952 → 1234 [PSH, ACK] Seq=19 Ack=
13 94.192682697 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=34 Win=
16 100.393560098 10.0.1.1 10.0.2.2 TCP 90 64 33952 → 1234 [PSH, ACK] Seq=34 Ack=
17 100.413686927 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=58 Win=
18 114.435244695 10.0.1.1 10.0.2.2 TCP 95 64 33952 → 1234 [PSH, ACK] Seq=58 Ack=
19 114.455399201 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [ACK] Seq=1 Ack=87 Win=
20 188.899288027 10.0.2.2 10.0.1.1 TCP 66 63 1234 → 33952 [FIN, ACK] Seq=1 Ack=87
21 188.902319253 10.0.1.1 10.0.2.2 TCP 66 64 33952 → 1234 [ACK] Seq=87 Ack=2 Win=
Frame 3: 74 bytes on wire (592 bits), 74 bytes captured (592 b
Ethernet II, Src: ea:29:b1:ed:75:ee (ea:29:b1:ed:75:ee), Dst:
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT
Total Length: 60
Identification: 0xb3dd (46045)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x6fdc [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.1.1
Destination Address: 10.0.2.2
Transmission Control Protocol, Src Port: 33952, Dst Port: 1234
0000 16 3d 92 a4 cf e4 ea 29 b1 ed 75 ee 08 00 45 00
0010 00 3c b3 dd 40 00 40 06 6f dc 0a 00 01 01 0a 00
0020 02 02 84 a0 04 d2 78 ed a5 f5 00 00 00 00 a0 02
0030 fa f0 17 31 00 00 02 04 05 b4 04 02 08 0a c2 3f
0040 ae 37 00 00 00 00 01 03 03 0a
Time to Live (ip.ttl), 1 byte(s)
Packets: 73 · Displayed: 13 (17.8%)
Profile: Default


Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT
Total Length: 60
Identification: 0xb3dd (46045)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x6fdc [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.1.1
Destination Address: 10.0.2.2
Transmission Control Protocol, Src Port: 33952, Dst Port: 1234
0020 02 02
0030 fa f0
0040 ae 37


cap2.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 1234
No. Time Source Destination Protocol Length Info
3 48.583717922 10.0.1.1 10.0.2.2 TCP 74 33952 → 1234 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SA
4 48.603977430 10.0.2.2 10.0.1.1 TCP 74 1234 → 33952 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0
5 48.604011757 10.0.1.1 10.0.2.2 TCP 66 33952 → 1234 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval
10 89.242792005 10.0.1.1 10.0.2.2 TCP 84 33952 → 1234 [PSH, ACK] Seq=1 Ack=1 Win=64512 Len=18
11 89.263041826 10.0.2.2 10.0.1.1 TCP 66 1234 → 33952 [ACK] Seq=1 Ack=19 Win=65536 Len=0 TSva
12 94.172398513 10.0.1.1 10.0.2.2 TCP 81 33952 → 1234 [PSH, ACK] Seq=19 Ack=1 Win=64512 Len=1
13 94.192682697 10.0.2.2 10.0.1.1 TCP 66 1234 → 33952 [ACK] Seq=1 Ack=34 Win=65536 Len=0 TSva
16 100.393560098 10.0.1.1 10.0.2.2 TCP 90 33952 → 1234 [PSH, ACK] Seq=34 Ack=1 Win=64512 Len=2
17 100.413686927 10.0.2.2 10.0.1.1 TCP 66 1234 → 33952 [ACK] Seq=1 Ack=58 Win=65536 Len=0 TSva
18 114.435244695 10.0.1.1 10.0.2.2 TCP 95 33952 → 1234 [PSH, ACK] Seq=58 Ack=1 Win=64512 Len=2
19 114.455399201 10.0.2.2 10.0.1.1 TCP 66 1234 → 33952 [ACK] Seq=1 Ack=87 Win=65536 Len=0 TSva
20 188.899288027 10.0.2.2 10.0.1.1 TCP 66 1234 → 33952 [FIN, ACK] Seq=1 Ack=87 Win=65536 Len=0
21 188.902319253 10.0.1.1 10.0.2.2 TCP 66 33952 → 1234 [ACK] Seq=87 Ack=2 Win=64512 Len=0 TSva
Frame 3: 74 bytes on wire (592 bits), 74 bytes captured (592 b
Ethernet II, Src: ea:29:b1:ed:75:ee (ea:29:b1:ed:75:ee), Dst:
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 10.0.2.2
Transmission Control Protocol, Src Port: 33952, Dst Port: 1234
0000 16 3d 92 a4 cf e4 ea 29 b1 ed 75 ee 08 00 45 00
0010 00 3c b3 dd 40 00 40 06 6f dc 0a 00 01 01 0a 00
0020 02 02 84 a0 04 d2 78 ed a5 f5 00 00 00 00 a0 02
0030 fa f0 17 31 00 00 02 04 05 b4 04 02 08 0a c2 3f
0040 ae 37 00 00 00 00 01 03 03 0a
cap2.pcap
Packets: 73 · Displayed: 13 (17.8%)
Profile: Default


kruthi@LAPTOP-LP9QU3RR: ~
etting up libqt6widgets6t64:amd64 (6.4.2+dfsg-21.1build5) ...
etting up libqt6multimedia6:amd64 (6.4.2-11build3) ...
etting up qt6-qpa-plugins:amd64 (6.4.2+dfsg-21.1build5) ...
etting up libqt6opengl6t64:amd64 (6.4.2+dfsg-21.1build5) ...
etting up tshark (4.2.2-1.1build3) ...
etting up libqt6svg6:amd64 (6.4.2-4ubuntu3) ...
etting up libqt6waylandclient6:amd64 (6.4.2-5build3) ...
etting up qt6-gtk-platformtheme:amd64 (6.4.2+dfsg-21.1build5) ...
etting up libqt6printsupport6t64:amd64 (6.4.2+dfsg-21.1build5) ...
etting up libqt6quick6:amd64 (6.4.2+dfsg-4build3) ...
etting up libqt6wlshellintegration6:amd64 (6.4.2-5build3) ...
etting up wireshark (4.2.2-1.1build3) ...
etting up libqt6waylandcompositor6:amd64 (6.4.2-5build3) ...
etting up libqt6waylandeglcompositorhwintegration6:amd64 (6.4.2-5build3) ...
etting up libqt6waylandeglclienthwintegration6:amd64 (6.4.2-5build3) ...
etting up qt6-wayland:amd64 (6.4.2-5build3) ...
rocessing triggers for libc-bin (2.39-0ubuntu8.9) ...
rocessing triggers for man-db (2.12.0-4build2) ...
rocessing triggers for shared-mime-info (2.4-4) ...
rocessing triggers for udev (255.4-1ubuntu8.17) ...
rocessing triggers for hicolor-icon-theme (0.17-2) ...
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add host1
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add router
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns add host2
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns list
ost2
outer
ost1
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-h1 type veth peer name veth-r1
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip link add veth-r2 type veth peer name veth-h2
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h1 netns host1
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r1 netns router
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-r2 netns router
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip link set veth-h2 netns host2
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set veth-h1 up
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip link set lo up
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set veth-r1 up
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set veth-r2 up
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router ip link set lo up
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip link set veth-h2 up
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip link set lo up
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
et.ipv4.ip_forward = 1
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
ING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
4 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=1.25 ms
4 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.045 ms
4 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.038 ms
-- 10.0.2.2 ping statistics ---
packets transmitted, 3 received, 0% packet loss, time 2196ms
tt min/avg/max/mdev = 0.038/0.443/1.247/0.568 ms
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router tc qdisc show dev veth-r2
disc noqueue 0: root refcnt 2
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%
ruthi@LAPTOP-LP9QU3RR:~$ ^C
ruthi@LAPTOP-LP9QU3RR:~$ sudo ip netns exec router tc qdisc show dev veth-r2
disc netem 8001: root refcnt 9 limit 1000 delay 20ms loss 1%
ruthi@LAPTOP-LP9QU3RR:~$